

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTEEC702 - POWER SYSTEM OPERATION & CONTROL

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following reactive power compensation technique is used during no load:
 - a. Shunt Capacitor
 - b. Shunt reactor
 - c. Series capacitor
 - d. none of the above
 - 2 Which of the following is used to boost local voltage?
 - a) Shunt Capacitor
 - b) Series capacitor
 - c) Synchronous condenser
 - d) None
 - 3 Which of the following compensation method is suitable in both no load condition as well as full load condition
 - a) Static VAR compensator
 - b) Shunt capacitor
 - c) Series capacitor
 - d) None
 - 4 Power system stability refers to the ability of a power system to
 - a) Supply uninterrupted power
 - b) Maintain voltage at all buses
 - c) Return to equilibrium point after a disturbance occur
 - d) Maintain constant frequency
 - 5 Transient stability deals with system behavior following
 - a) Small load changes
 - b) Gradual voltage variations
 - c) Large sudden disturbances
 - d) Frequency regulation

- 6 Increasing excitation voltage of generator
 - a) Reduces stability
 - b) Improves steady-state stability
 - c) Has no effect
 - d) Causes oscillations
- 7 Increase in excitation of a synchronous generator will
 - a) Decrease terminal voltage
 - b) Increase terminal voltage
 - c) Reduce frequency
 - d) Reduce speed
- 8 Which of the following is a DC excitation system?
 - a) Brushless excitation
 - b) Static excitation
 - c) Rotating DC exciter
 - d) AVR-based excitation
- 9 The main components of an AVR system include
 - a) Governor, turbine, generator
 - b) Amplifier, exciter, generator, sensor
 - c) Transformer, relay, breaker
 - d) Governor, load, turbine
- 10 Frequency deviation in a power system is mainly caused by imbalance between
 - a) Voltage and current
 - b) Reactive power and voltage
 - c) Mechanical power and electrical load
 - d) Excitation and speed
- 11 The fuel cost curve of a generator is generally
 - a) Linear
 - b) Exponential
 - c) Quadratic
 - d) Sinusoidal
- 12 Economic dispatch determines
 - a) Which generators should be ON or OFF
 - b) Optimal power output of committed generators
 - c) Location of generating stations
 - d) Voltage levels of generators

Q2. Solve the following.

- A) What is the need of reactive power compensation? Explain a static VAR compensator (SVC) with help of circuit diagram and V-I characteristics? Also, Discuss the ability of SVC to handle both light load condition and full load condition? 6
- B) Explain the working of Synchronous condenser with the help of neat and clean diagram? Also, Discuss the advantages synchronous condensor offer when compared to other reactive power control techniques. 6

Q3. Solve the following.

- A) Explain different elements of excitation system with help of neat and clean diagram? 6
- B) List out the types of excitation systems? Explain any one type of AC excitation system with the help of a neat and clean diagram? 6

Q4. Solve Any Two of the following.

- A) A 50Hz, 100-MVA, four pole, synchronous generator has an inertia constant of 3.5s and is supplying 0.16p.u power on a system base of 500MVA. The input to the governor is increased to 0.18p.u. Determine i) the kinetic energy stored in the moving parts of the generator and ii) the acceleration of the generator. If the acceleration continues for 7.5 cycles, calculates iii) the change in rotor angle and iv) the speed in RPM at the end of the acceleration. 6
- B) Derive swing equation of a synchronous generator? How to calculate mechanical starting time and inertia constant? 6
- C) Explain the methods employed to improve Transient stability of power system? 6
- Q5. Solve Any Two of the following.
- A) Draw the schematic diagram of Automatic voltage regulator (AVR) loop. Derive the transfer function of the AVR loop? 6
- B) Explain the equal area criterion in detail and discuss any one of its applications. 6
- C) Discuss the working of speed governing system with proper diagram? 6
- Q6. Solve Any Two of the following.
- A) The fuel inputs per hour of plants 1 and 2 are given as 6
- $$F_1 = 0.2 P_1 + 40 P_1^2 + 120 \text{ Rs. per hr}$$
- $$F_2 = 0.25 P_2 + 30 P_2^2 + 150 \text{ Rs. per hr}$$
- Determine the economic operating schedule and the corresponding cost of generation if the maximum and minimum loading on each unit is 100 MW and 25 MW, the demand is 180 MW, and transmission losses are neglected. If the load is equally shared by both the units, determine the saving obtained by loading the units as per equal incremental production cost
- B) Write a short note on Unit Commitment, including its objectives, significance, and basic approach. 6
- C) Explain the load frequency control (LFC) of a single-area power system in detail with the help of a neat block diagram 6

***** End *****